OPEN SOURCE APACHE 2.0

# MCP Gateway & AI Registry

Unified Control Plane for AI Agents, MCP Servers, and Agent Skills

MCP SERVERS A2A AGENTS AGENT SKILLS ENTERPRISE READY

GitHub Repository AWS Workshop

Last updated: April 11, 2026

# The AI Asset Sprawl Problem

### Without a Control Plane

- ✗ Separate connections per MCP server
- ✗ Each developer configures tools individually
- ✗ Scattered API keys and credentials
- ✗ No visibility into tool usage
- ✗ No dynamic tool discovery for agents
- ✗ Separate agent and server governance

### With Gateway & Registry

- ✓ One gateway, access all MCP servers
- ✓ Single config for VS Code, Cursor, Claude Code
- ✓ Centralized credential management
- ✓ Complete audit trails and analytics
- ✓ Semantic discovery for servers, agents, skills
- ✓ Unified governance for all AI assets

**A central control plane for all your AI assets -- servers, agents, and skills**

# Four Core Functions

### Unified MCP Gateway

Centralized access point for multiple MCP servers with reverse proxy routing, SSL termination, and health monitoring.

### MCP Server Registry

Register, discover, and manage MCP servers with unified governance, semantic search, and version routing.

### Agent Registry & A2A Hub

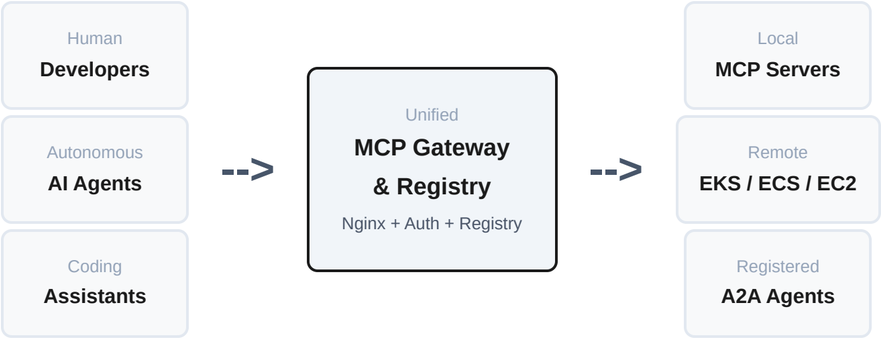
Agent registration, discovery, and direct agent-to-agent communication via the A2A Protocol. Supports both A2A and non-A2A agents.

### Skills Registry

Register, version, and discover reusable AI skills (SKILL.md) with security scanning, enabling skill sharing across agents and teams.

# Platform Demo

# Platform Architecture




Single connection point with **dual authentication** supporting both human users and machine-to-machine agent flows.

# Enterprise Security

### Multi-Provider IAM
Keycloak, Microsoft Entra ID, Okta, Auth0, and AWS Cognito with harmonized management API.

### Fine-Grained Access
Scopes define which servers, methods, tools, and agents each user or service account can access.

### Security Scanning
Cisco AI Defense integration for MCP servers and A2A agents with YARA pattern matching.

### Audit Logging
Comprehensive audit trails with credential masking, TTL retention, and SOC 2/GDPR compliance.

### M2M Auth
OAuth2 Client Credentials flow for AI agent identity with service accounts.

### Static Token Auth
API key access for CI/CD pipelines and trusted network environments without IdP setup.

# Intelligent Discovery

### Hybrid Search

- Vector similarity + keyword matching
- Unified search across servers, tools, agents, skills
- Name-boost for exact match relevance
- Tag-based and metadata filtering

### Flexible Embeddings

- Local sentence-transformers
- OpenAI embeddings
- Amazon Bedrock Titan
- 100+ LiteLLM-supported providers

Agents discover tools at runtime via natural language -- **"find an agent that can book flights"**

# Agent Registry & A2A Protocol

### Agent-to-Agent Communication

Direct peer-to-peer agent communication. Registry handles discovery, agents connect directly for low-latency interaction.

### Semantic Agent Discovery

Agents find collaborators via natural language queries. Dynamic composition based on capabilities, not hardcoded references.

### Agent Skills Registry

Register reusable SKILL.md instruction sets with security scanning, visibility controls, and star ratings.

Agents autonomously discover specialized agents for tasks they cannot handle, enabling **dynamic agent orchestration**.

# Agent Trust: Agent Name Service

**DNS + PKI for AI Agents** -- just as SSL certificates let you trust websites, ANS lets you trust agents.

### Structured Identity

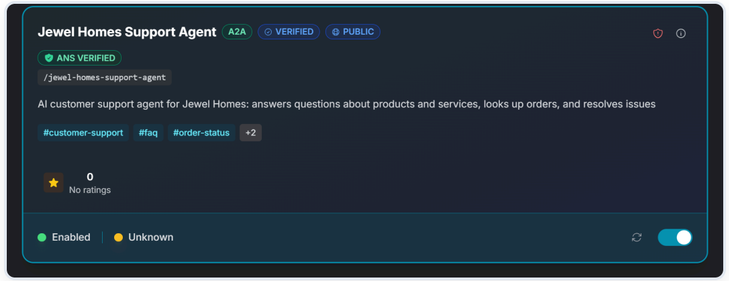
Each agent gets an ANS Name and X.509 certificate issued by a Registration Authority.

### Bring Your Own ANS ID

Owners link their ANS Agent ID to registry entries. Registry verifies via GoDaddy ANS API.

### Production Resilience

Circuit breaker, retry with backoff, 6-hour background re-verification, and rate limiting.




Verified agents display a trust badge with certificate details

**IETF Internet-Draft** (draft-narajala-ans-00) -- Powered by GoDaddy ANS, integrated into the open-source AI Registry

# Registry Federation

### P2P Federation

Bidirectional sync between registry instances with configurable sync modes: all, whitelist, or tag filter.

### Anthropic Registry

Import curated MCP servers from Anthropic's official registry with full API compatibility.

### External Sources

AWS Agent Registry, Workday ASOR, and Anthropic Registry -- federate assets from multiple external sources.

### Registry Card

Standardized discovery via /.well-known/registry-card with capabilities, auth endpoints, and contact info.

Central IT aggregates across LOB registries -- LOBs inherit shared tools from a **central hub**.

# AWS Agent Registry Federation

**Federate MCP servers, A2A agents, and skills** from AWS Agent Registry into the Gateway.

### Multi-Registry Support

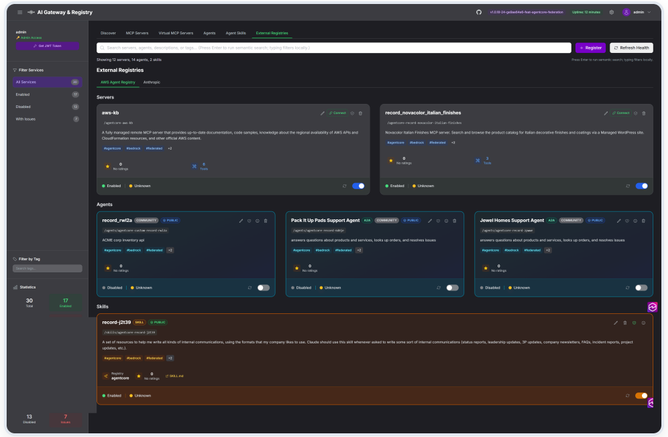
Add multiple registries from same or different AWS accounts and regions. Cross-account via IAM role assumption.

### Selective Sync

Choose descriptor types (MCP, A2A, CUSTOM, AGENT_SKILLS) and status filters per registry.

### Cascade Cleanup

Remove a registry and all its synced assets are automatically deregistered.




Federated assets from AWS Agent Registry in the External Registries tab

Enable with a single env var: **AWS_REGISTRY_FEDERATION_ENABLED=true** -- manage registries via UI, API, or CLI

# Virtual Servers & Version Routing

## Virtual MCP Servers

- Aggregate tools from multiple backends
- Tool aliasing for naming conflicts
- Per-tool scope-based access control
- Session multiplexing (1:N transparent)
- 60s cached aggregation

## Version Routing

- Multiple versions behind single endpoint
- Test with X-MCP-Server-Version header
- Instant rollback with one API call
- Deprecation lifecycle with sunset dates
- O(1) nginx map-based routing

# Observability & Metrics

### Grafana Dashboards

Real-time metrics for server health, tool usage, auth events, and system performance.

### OTLP Push Export

Direct export to Datadog, New Relic, Honeycomb, or Grafana Cloud -- no OTEL Collector needed.

### OpenTelemetry

Integration with Prometheus, CloudWatch, and other monitoring platforms via OTEL protocol.

### Anonymous Telemetry

Privacy-first usage telemetry for adoption tracking. Non-sensitive metadata only. Opt-out with one env var.

# Flexible Deployment

### Amazon ECS Fargate

Serverless containers with multi-AZ, ALB with HTTPS, auto-scaling, CloudWatch, and NAT Gateway HA. Full Terraform config.

### Amazon EKS (Kubernetes)

Helm chart deployment on Amazon EKS for container orchestration at scale.

### Amazon EC2

Traditional deployment on EC2 instances with Docker Compose. Complete setup guide from scratch.

### Docker / Podman

Pre-built Docker Hub images and rootless Podman support for macOS and Linux. One-command AI-assisted setup.

**3**
AWS Deployment Targets

**5**
Identity Providers

**700+**
Passing Tests

# Developer Experience

### MCP Registry CLI

Conversational interface for registry management with real-time token status and cost tracking.

### IAM Settings UI

Visual management of users, groups, and M2M service accounts with fine-grained permissions.

### AI Coding Assistants

Single config for VS Code, Cursor, Claude Code, and Cline with guided tool discovery.

### Management API

RESTful HTTP endpoints with Python client, type-safe interfaces, and comprehensive error handling.

### Config Viewer

Admin panel with 11 config groups. Export as ENV, JSON, TFVARS, or YAML for deployment automation.

### Custom Metadata

Rich metadata on servers and agents for team ownership, compliance tracking, and cost allocation.

# What's New

- **AWS Agent Registry Federation**
  Federate MCP servers, A2A agents, and skills from AWS Agent Registry. Multi-registry, cross-account, cascade cleanup.
- **Register Any Agent (A2A and Non-A2A)**
  New supported_protocol field distinguishes agent types. Register through UI, API, or CLI.
- **Amazon Bedrock AgentCore Bulk Import**
  Auto-discover and register all AgentCore Gateways and Agent Runtimes from your AWS account.
- **Agent Name Service (ANS) Integration**
  PKI-based trust verification for agents via GoDaddy ANS. Green shield badge on verified agents.
- **Auth0 Identity Provider Support**
  Fifth IdP alongside Keycloak, Entra ID, Okta, and Cognito. Harmonized IAM API for all providers.
- **Anonymous Usage Telemetry**
  Privacy-first deployment metadata. No PII. Opt-out with MCP_TELEMETRY_DISABLED=1.
- **Registry Card for Federation Discovery**
  Standardized /.well-known/registry-card with capabilities and auth metadata.
- **AWS Workshop Studio**
  Hands-on workshop for deploying and securing AI agent ecosystems on AWS.
- **AI Registry MCP Server (airegistry-tools)**
  5 tools for AI coding assistants to discover and query registry directly.

# What's Next

**Registry Copilot** An AI assistant that helps you build, register, and configure your agents through natural language

**Enterprise Readiness** Enhanced UI dashboards, structured audit logging, RBAC refinements, and advanced security controls

**Have a feature idea?**

We build in the open. Tell us what you need.

**Create an issue on GitHub →**

# Demo & Links

### AWS Show & Tell

YouTube -- MCP Gateway & Registry deep dive

### Live Demo

Vidcast -- Hands-on walkthrough of the platform

### Roadmap

GitHub Milestones -- Upcoming features and releases

# One AI Registry. Unified Control.

**3**
**AWS Targets**
ECS Fargate, EKS, EC2

**5**
**Identity Providers**
Keycloak, Entra ID, Okta, Auth0, Cognito

**Open**
**Source**
Apache 2.0 License

github.com/agentic-community/mcp-gateway-registry

**Quick Start** | **Documentation** | **AWS Deployment**

**AWS Workshop** | **Demo Video** | **Community**